**8** (a) Work out an estimate for the value of $\sqrt{63.5 \times 101.7}$

(a) $63.5 \approx 64$; $101.7 \approx 100$

$\sqrt{(63.5 \times 101.7)} \approx \sqrt{(64 \times 100)}$

$= \sqrt{6400} = 80$

**1 mark**
Sensible rounded values.

**1 mark**
Estimate in range 75–81.

**(2)**

$(2.3)^6 = 148$ correct to 3 significant figures.

(b) Find the value of $(0.23)^6$ correct to 3 significant figures.

(b) 0.23 is one tenth of 2.3.

$(0.23)^6 = (2.3)^6 \div 10^6$

$= 148 \div 1\,000\,000 = 0.000148$

**1 mark**
0.000148 to 3 significant figures.

**(1)**

(c) Find the value of $5^{-2}$

(c) $5^{-2} = 1 \div 5^2 = 1/25$

**1 mark**
1/25, or 0.04.

**(1)**

**(Total for Question 8 is 4 marks)**